

Concurrency and Distributed Systems

Week 10 – Part 2: Queue to Socket Protocol

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Part 2 Roadmap

1 Boundary

2 Protocol

3 Demo

Local Queue vs Socket Queue

Model	Shared object?	Failure mode
BlockingQueue	yes, in process	waiting, interruption
Socket protocol	no	latency, timeout, reset

Simple Protocol

```
PUT <payload>  
TAKE  
ACK <payload>  
EMPTY
```

Meaning

We move queue semantics into explicit messages.

Design Questions

- How does the client know the server accepted a message?
- What happens when the server is slow?
- What happens when the connection resets?
- Is retry safe for this command?

Run

```
.\run-demo.ps1 02
```

```
.\run-demo.ps1 03
```